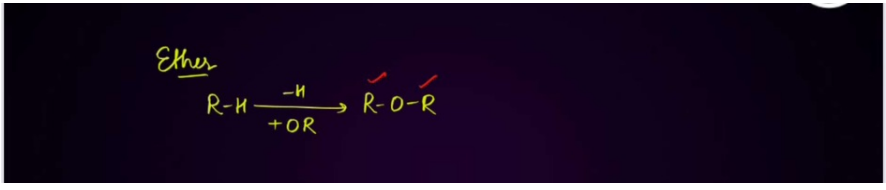
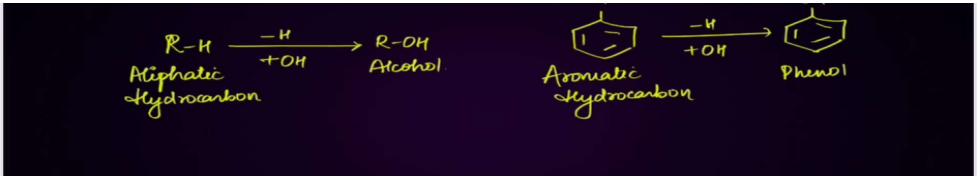
12th hacker series

Alcohol , phenol and ether

Introduction

- alcohols- hydroxy, derivatives of hydrocarbons are called alcohols. They are formed when a hydrogen atom is replaced by OH group.
- phenols- aromatic hydroxy compounds in which OH group is bounded to benzene ring directly are called phenols
- Ether- the substitution of H in a hydrocarbon by an OR group gives ether



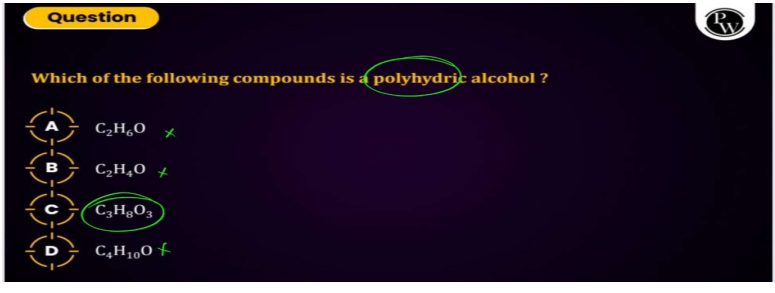


Classification of alcohol

On the basis of number of OH group present

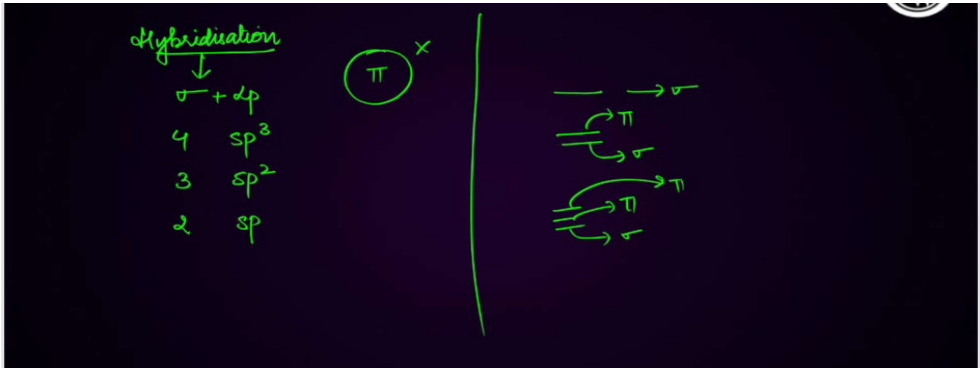
1) monohydric alcohol → 1 OH group , eg: CH_3CH_2OH
2) dihydric alcohol → 2 OH group , eg: $CH_3-\underset{OH}{CH}-\underset{OH}{CH_2}$
3) trihydric alcohol → 3 OH group

eg- $CH_2(OH)-CH(OH)-CH_2(OH)$



On the basis of hybridisation of carbon atom to which OH is attached

- $C_{sp^3}-OH$
  - → degree of alcohol
  - → allylic alcohol
  - → benzylic alcohol
- $C_{sp^2}-OH$
  - → vinylic alcohol
  - → phenol



$C_{sp^3}-OH$

i) degree of alcohol

- 1° alcohol
  - $1^\circ C-OH$
  - eg CH_3-CH_2-OH
- 2° alcohol
  - $2^\circ C-OH$
  - eg $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH$
- 3° alcohol
  - $3^\circ C-OH$
  - eg $CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}}-OH$

(ii) Allylic Alcohol

$C=C-\overset{\text{allylic carbon}}{C}-OH$

eg – $CH_2=CH-CH_2-OH$

1° allylic alcohol

$H_2C=CH-\underset{1C}{CH_2}-OH$

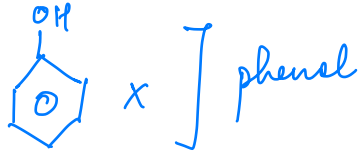
2° allylic alcohol

$H_2C=CH-\underset{2C}{CH}(CH_3)-OH$

3° allylic alcohol

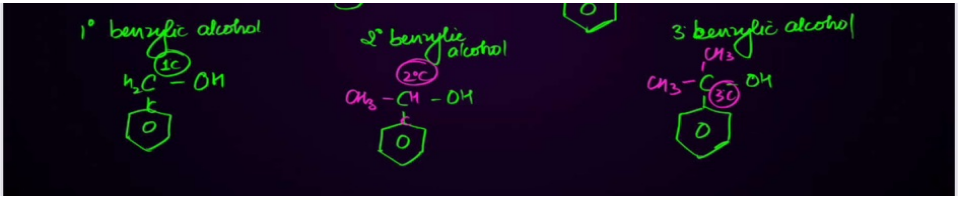
$H_2C=CH-\underset{3C}{C}(CH_3)_2-OH$

(iii) Benzylic alcohol



eg –





Question

In allylic and benzylic alcohol ,-OH group is attached to-

$C_{sp^3}-OH$

A. Sp^3 , sp^3 hybridized carbon atom

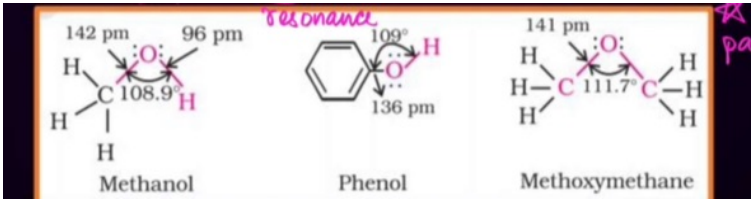
B. Sp^2 , sp^3 hybridized carbon atom

C. Sp , sp^2 hybridized carbon atom

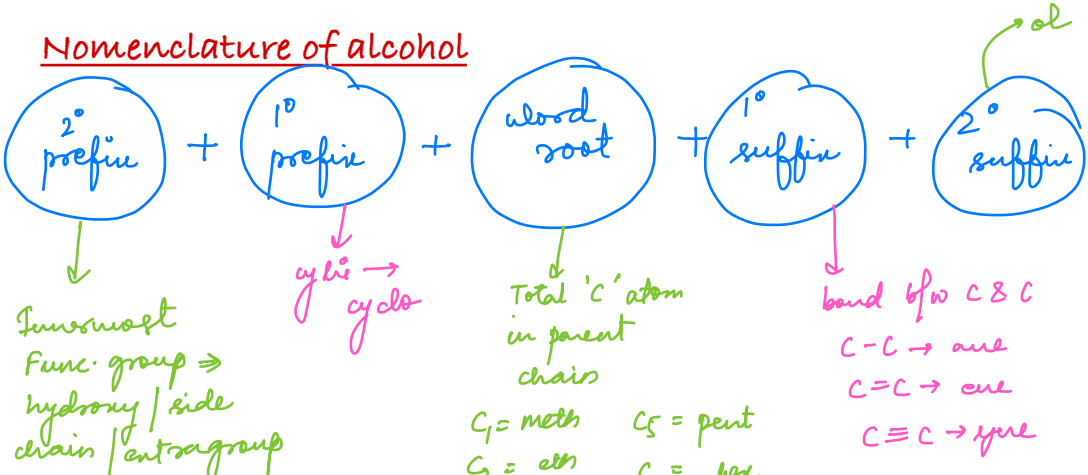
D. Sp^2 , sp hybridized carbon atom

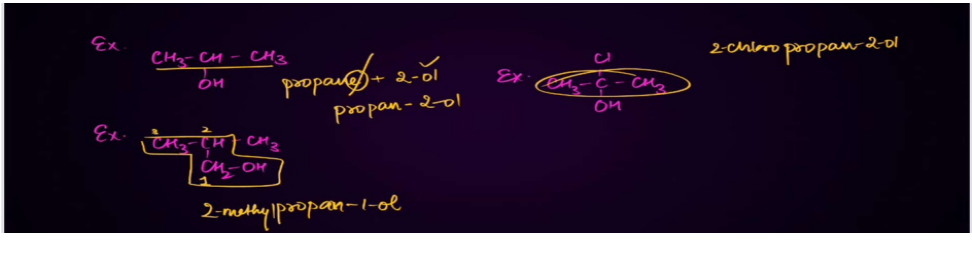
Structure of functional group

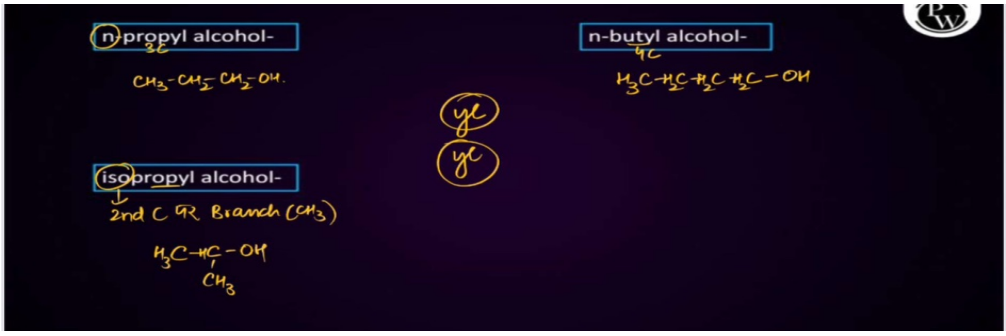
- in alcohols, the oxygen of the OH group is attached to the carbon by a sigma bond formed by the overlap of the SP3 hybridised orbital of the carbon with the SP3 hybridised orbital of oxygen



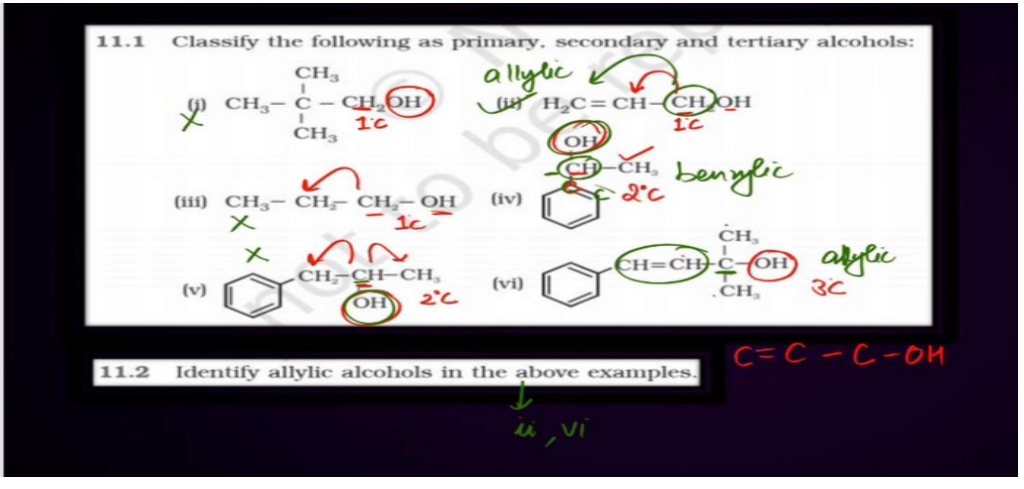
Nomenclature of alcohol











MOP of alcohol

1) Alkyl halide

$$R-X \xrightarrow[SN_2]{aq.\ KOH} R-OH + KX$$

$$CH_3CH_2Cl \xrightarrow[SN_1]{H_2O} CH_3CH_2OH + HCl$$

2) hydration of alkenes

Direct hydration
→ Acid catalysed hydration
H^+ / H_2O
markovnikov

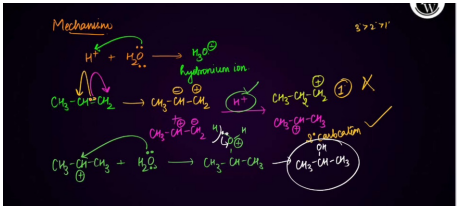
Indirect hydration

Hydroboration oxidation
BH_3 or B_2H_6 - THF / H_2O_2
Anti markovnikov

Oxymercuration - demercuration
Hg
markovnikov

eg

$CH_2 = CH_2 \xrightarrow{H^+/H_2O} CH_3CH_2OH$



eg –

$CH_3-CH=CH_2 \xrightarrow[H_2O_2]{BH_3 \text{ or } B_2H_6\text{-THF}} CH_3CH_2CH_2OH$

eg

$CH_3-CH=CH_2 \xrightarrow[H_2O]{Hg(OAc)_2,\ NaBH_4} CH_3-\underset{OH}{CH}-CH_3$

3) Grignard reagent

$R-MgX + R-\overset{O}{\overset{\|}{C}}-R \longrightarrow R-\underset{R}{\overset{O^-Mg^+X}{\overset{|}{C}}}-R \xrightarrow{H^+/OH^-} R-\underset{R}{\overset{OH}{\overset{|}{C}}}-R + Mg(OH)X$

4) carbonyl compounds

- by reduction of aldehydes and ketones:aldehydes and ketones are reduced to the corresponding alcohol by addition of hydrogen in presence of catalyst(catalytic hydrogenation). The usual catalyst is finally divided in metal such as platinum, palladium or nickel. It is also prepared by treating aldehydes and ketones with sodium borohydride or lithium aluminium hydride. Aldehyde yield primary alcohol, whereas ketone gives secondary alcohol.
- by reduction of carboxylic acids and Esters: carboxylic acids are reduced to primary alcohol is in excellent yields by lithium aluminium hydride, a strong reducing agent

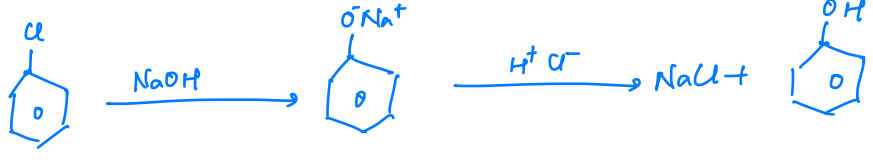
① $R-\overset{O}{\overset{\|}{C}}-H \xrightarrow[NaBH_4 \text{ / } Ni, Pt, Pd, H_2]{LiAlH_4} R-CH_2-OH$

② $R-\overset{O}{\overset{\|}{C}}-H \xrightarrow{LiAlH_4} RCH_2OH$

MOP of phenol

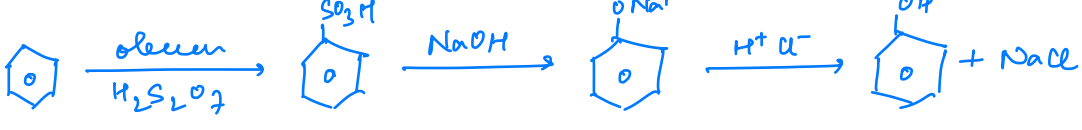
1) from haloarenes

Chlorobenzene is fused with NAOH at 623 Kelvin and 320 atmospheric pressure. Phenol is obtained by acidification of sodium phenoxide so produced.



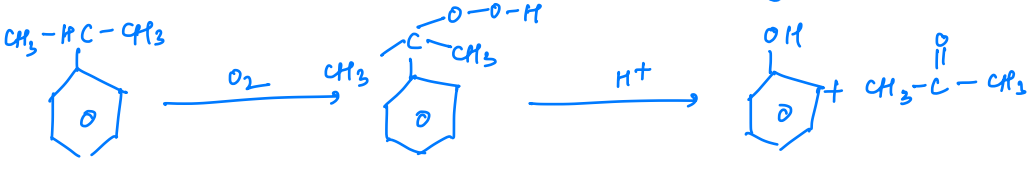
2) from benzenesulphonic acid

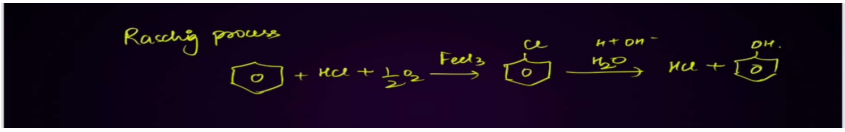
Benzene is sulphonated with oleum and benzene sulphonic acid so formed is converted to sodium phenoxide on heating with molten sodium hydroxide. Acidification of sodium salt gives phenol.



3) from cumene

Phenol is manufactured from the hydrocarbon, cumene. Cumene is oxidised in presence of air to cumene hydroperoxide. It is converted to phenol and acetone by treating it with dilute acid acetone by product of this reaction is also obtained in large quantities by this method.









MOP of ether

1) Williamson synthesis

$R^+X^- + R-\bar{O}-\overset{+}{N}a \longrightarrow NaX + R-O-R$

ether

2) dehydration of alcohol

$CH_3CH_2OH \xrightarrow[413K]{140^\circ C} CH_3CH_2OCH_2CH_3$

ether

$170^\circ C$ 443K

Alkene

$CH_2 = CH_2$

Question

What would be the reactant and reagent used to obtain 2,4-dimethylpentan-3-ol?

A. Propanal and propyl magnesium bromide

B. 3-Methylbutanal and 2-methyl magnesium iodide

C. 2, 2-Dimethylpropanone and methyl magnesium iodide

D. 2-Methylpropanal and isopropyl magnesium iodide

Physical properties of alcohol

- boiling point- $\propto$ Molecular mass
  $\propto \frac{1}{\text{Branching}}$
- solubility- lower molecule soluble in water, number of carbon atom increases, hydrophobic part increases, solubility decreases

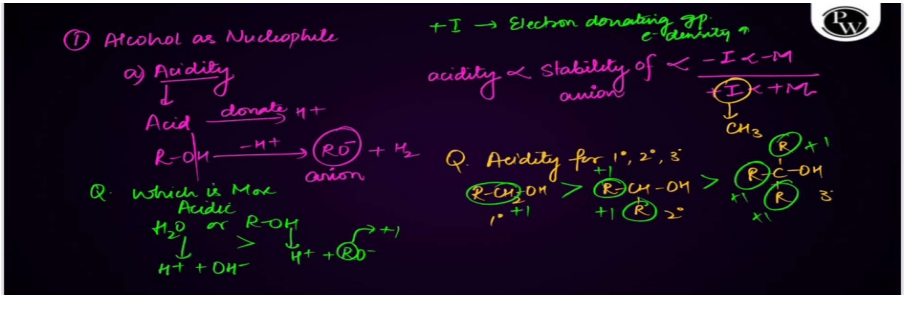
Chemical properties of alcohol

As a NUCLEOPHILE

$R-O\overset{\ominus}{}H$

As a ELECTROPHILE

$R\overset{\oplus}{}O-H$



b) Rx with Metals → K/Ac.

$R\text{-}OH + Na \longrightarrow R\text{-}O\text{-}Na + H_2$

c) Esterification

↓

Formation of Ester

$R\text{-}OH + R\text{-}\overset{O}{\overset{\|}{C}}\text{-}OH \longrightarrow H_2O + R\text{-}\overset{O}{\overset{\|}{C}}\text{-}OR$

QUESTION

Assertion (A): Ortho and para-nitrophenols can be separated by steam distillation.
Reason (R): Ortho isomer associates through intermolecular hydrogen bonding while Para isomer associates through intramolecular hydrogen bonding.

[CBSE, 2020]

Solution

Due to intermolecular H-bonding in para nitrophenol, para nitrophenol has a more boiling point than ortho nitrophenol. Therefore, they can be separated by steam distillation.

QUESTION

Assertion (A): $(CH_3)_3COCH_3$ gives $(CH_3)_3C\text{–}I$ and CH_3OH on treatment with HI.
Reason (R): The reaction occurs by S_N1 mechanism.

[CBSC, 2020]

Solution

This reaction proceeds by S_N1 mechanism as the formation of tertiary carbocation takes place.

QUESTION

Arrange the following in order of increasing boiling point:
Ethoxyethane, Butanal, Butanol, n-butane

ether aldehyde alcohol hydrocarbon

I II III IV

IV < I < II < III

QUESTION

What happens when
(a) Anisole is treated with CH_3Cl/ anhydrous $AlCl_3$?
(b) Phenol is oxidised with $Na_2Cr_2O_7//H^+$?
(c) $(CH_3)_3C - OH$ is heated with Cu/573K?
Write a chemical equation in support of your answer.

Solution

(a) Anisole (OCH_3) + CH_3Cl (anhydrous $AlCl_3$) → o-methylanisole (OCH_3, CH_3) + p-methylanisole (OCH_3, CH_3)

(b) Phenol (OH) + $Na_2Cr_2O_7$, H_2SO_4 → Benzoquinone

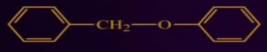
(c) $CH_3\text{–}C(CH_3)_2\text{–}OH \xrightarrow{Cu/573K} CH_3\text{–}C(CH_3)=CH_2$

QUESTION

Give the structures of final products expected from the following reactions: [CBSE 2020]

(a) Hydroboration of propene followed by oxidation with H_2O_2 in alkaline medium.

(b) Dehydration of $(CH_3)_3C - OH$ by heating it with 20% H_3PO_4 at 358 K.

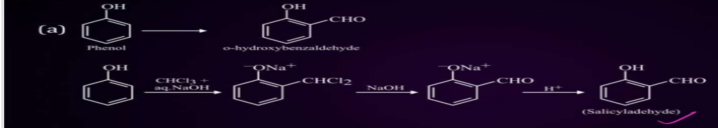
(c) Heating of  with HI.

QUESTION

Convert:

(a) Phenol to o-hydroxybenzaldehyde

Solution

(a) 

QUESTION

Convert:

(b) Methanal to ethanol

Solution

(b) $\underset{\text{Methanal}}{HCHO} \rightarrow \underset{\text{Ethanol}}{CH_3CH_2OH}$

$$H-\overset{O}{\overset{\|}{C}}-H + CH_3MgBr \rightarrow H_3C-CH_2-OMgBr \xrightarrow{H_2O} CH_3CH_2OH + Mg(OH)Br$$

QUESTION

Convert:

(c) Phenol to phenyl ethanoate.

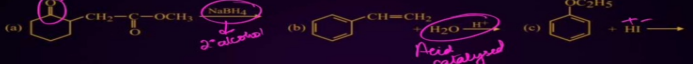
Solution

(c) 

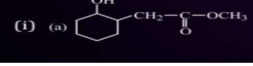
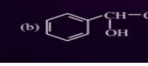
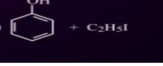


QUESTION

(i) Write the structures of the main products in the following reactions:



Solution

(i) (a) 
(b) 
(c)  $+ C_2H_5I$

QUESTION

(i) (a) Ortho-nitrophenol is more acidic than ortho-methoxyphenol. Give reason.

[CBSE, 2019]

(b) Draw the structure of the major monohalo product in the following reaction:



QUESTION

(a) When 3-methylbutan-2-ol is treated with HBr, the following reaction takes place:



Give a mechanism for this reaction.
(Hint: The secondary carbocation formed in step II rearranges to a more stable tertiary carbocation by a hydride ion shift from 3rd carbon atom.)

Solution

(a) **Step 1:** Protonation



Step 2: Formation of 2° carbocation by the elimination of a water molecule



Step 3: Rearrangement by the hydride-ion shift



Step 4: Nucleophilic attack



QUESTION

Which type of reaction is used in Williamson's synthesis of ethers?

Solution

S_N2 types of reaction is occur in Williamson synthesis.

QUESTION

What is the role of protic acid in dehydration of ethers?

Solution

Polar acid helps to protonate alcohol, after this, removal of water will take place.

QUESTION

How does the presence of alkoxy groups in aromatic ethers affect their reactivity towards electrophilic substitution reactions?

Solution

Aromatic ether containing alkoxy groups activates their aromatic sings towards Electrophilic substitution reactions. This makes them more susceptible to reaction like halogenation, nitration etc.

QUESTION



Solution



QUESTION

Which of the following alcohols is resistant to oxidation?

(A) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH$

(B) $CH_3-\underset{\underset{CH_3}{|}}{CH}-OH$

(C) CH_3-CH_2-OH

(D) CH_3-OH

Solution

(A)
Tertiary alcohols are resistant to oxidation because the C-atom to which -OH group is attached does not have a hydrogen atom.

QUESTION

Which of the following groups increase the acidic character of phenol?

(A) CH_3O^-

(B) CH_3^-

(C) NO_2^-

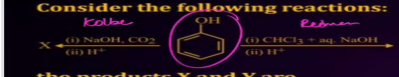
(D) All of these

Solution

(C)
Nitro group is an electron withdrawing group, it increases the acidic strength of phenol.

QUESTION

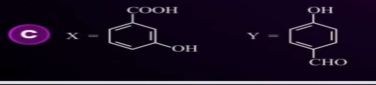
Consider the following reactions:



the products X and Y are

(A) 

(B) 

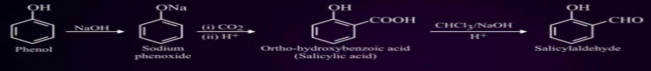
(C) 

(D) 

Solution

(D)

Product "X" is Salicylic acid and the reaction is Kolbe's reaction.



Thank you!!